ECS task

Installed Docker on the EC2 instance and pull the hiring application image for the ECS high-availability deployment.

Successfully pulled the Docker image `sabair0509/hiring-app:works`.

```
[root@ip-172-31-46-224 ~]# docker pull sabair0509/hiring-app:works
works: Pulling from sabair0509/hiring-app
44ba2882f8eb: Pull complete
2cabec57fa36: Pull complete
c4e554d59e12: Pull complete
111cbdbeed15: Pull complete
ebe3e1fbfc69: Pull complete
dfa1fc9cfd4d: Pull complete
33eb4f163d71: Pull complete
064a77b1040e: Pull complete
3e172369cc21: Pull complete
Digest: sha256:5807e0f53355d31d2506e0b41a37d08a1ad33bc5c4395dddf5b13d065b2afdbb
Status: Downloaded newer image for sabair0509/hiring-app:works
docker.io/sabair0509/hiring-app:works
[root@ip-172-31-46-224 ~]# docker image inspect sabair0509/hiring-app:works \
  --format '{{json .Config.ExposedPorts}}'
{"8080/tcp":{}}
[root@ip-172-31-46-224 ~]#
```

Result: Docker is installed and running, and the hiring application image is available locally for the ECS deployment.

- Attached the required ECS EC2 container-instance policy.
- Verified the attached IAM policies through AWS CLI.
- Next: verify the EC2 trust relationship before configuring the cluster.

Installed terraform

```
A newer release of "Amazon Linux" is available.
  Version 2023.12.20261026:
Run "/usr/bin/dnf check-release-update" for full release and version update info
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct 24 16:37:34 2026 from 13.239.158.5
[ec2-user@ip-172-31-46-224 ~]$ sud su -
-bash: sud: command not found
[ec2-user@ip-172-31-46-224 ~]$ sudo su -
Last login: Sat Oct 24 16:37:42 UTC 2026 on pts/1
[root@ip-172-31-46-224 ~]# terraform --version
Terraform v1.17.0
on linux_amd64
```

ECS High Availability Cluster with ALB

Objective: To deploy a highly available application on Amazon ECS using a load balancer, dynamic port mapping, and multiple Availability Zones.

Implementation Steps:

1. Created the ECS cluster `ecs-ha-cluster`.
2. Configured an Application Load Balancer named `ecs-ha-alb`.
3. Created the target group `ecs-ha-targets`.
4. Deployed Docker image `sabair0509/hiring-app:works`.
5. Configured dynamic port mapping for ECS tasks.
6. Set the ECS service desired count to 6 tasks.
7. Configured the ALB health-check path as `/hiring/`.
8. Updated security-group rules to allow traffic from the ALB to ECS dynamic ports.
9. Verified 6 desired tasks, 6 running tasks, and 0 pending tasks.
10. Verified 6 healthy ALB target registrations.
11. Verified ECS container instances across Availability Zones `ap-southeast-2a`, `ap-southeast-2b`, and `ap-southeast-2c`.

Result: Six ECS tasks are running, six healthy target registrations were verified, and ECS container instances are distributed across three Availability Zones.

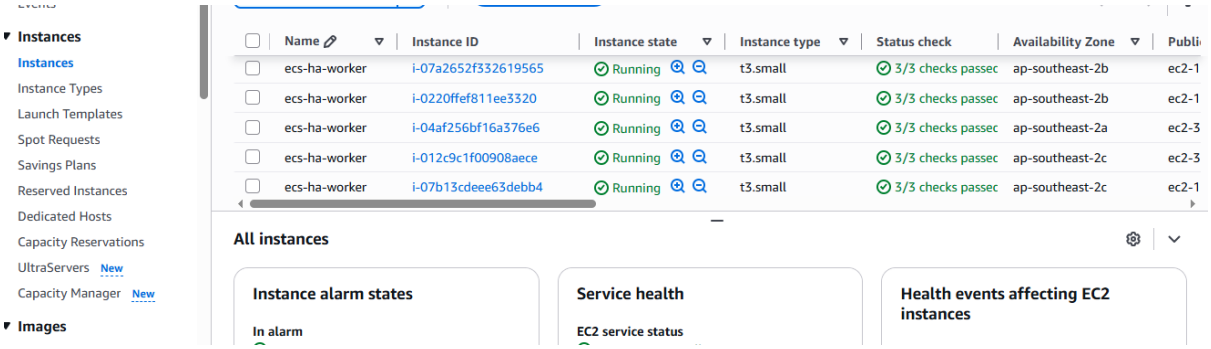
```
[root@ip-172-31-46-224 ~]# aws iam list-attached-role-policies \
  --role-name 'ecsInstanceRole.'
{
    "AttachedPolicies": [
        {
            "PolicyName": "AmazonEC2ContainerServiceRole",
            "PolicyArn": "arn:aws:iam::aws:policy/service-role/AmazonEC2ContainerServiceRole"
        },
        {
            "PolicyName": "AmazonEC2ContainerServiceforEC2Role",
            "PolicyArn": "arn:aws:iam::aws:policy/service-role/AmazonEC2ContainerServiceforEC2Role"
        }
    ]
}
[root@ip-172-31-46-224 ~]# aws iam get-role \
  --role-name 'ecsInstanceRole.' \
  --query 'Role.AssumeRolePolicyDocument' \
  --output json
{
    "Version": "2008-10-17",
    "Statement": [
        {
            "Sid": "",
            "Effect": "Allow",
            "Principal": {
                "Service": "ecs.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

Auto scaling

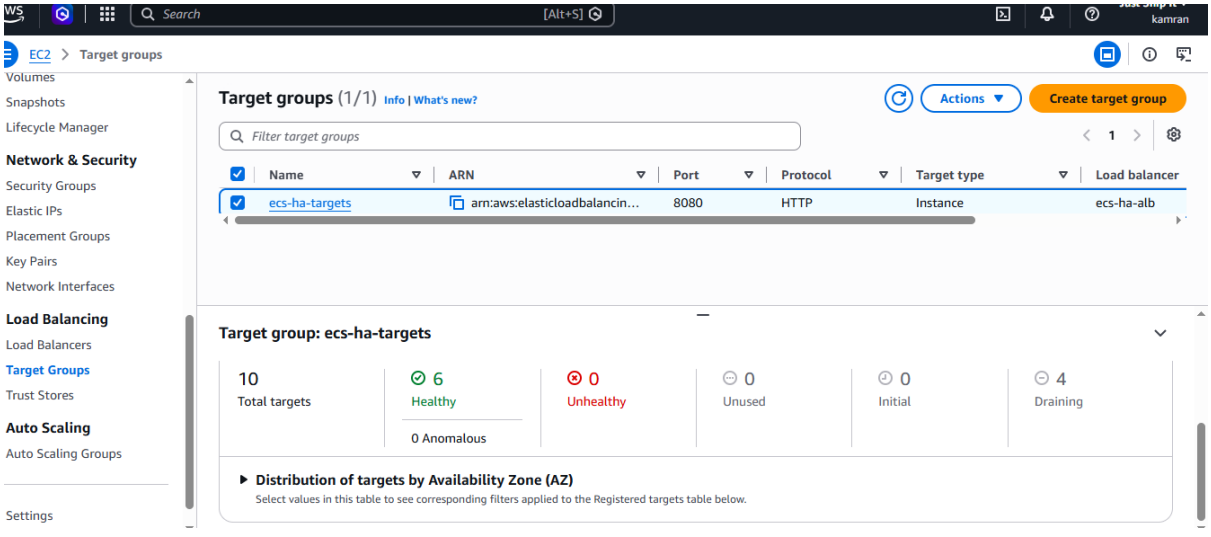
```
[root@ip-172-31-46-224 ecs-ha-project]# aws ec2 describe-instances \
  --region ap-southeast-2 \
  --filters Name=instance-state-name,Values=running,pending \
  --query 'Reservations[].Instances[].{ID:InstanceId,Type:InstanceType,State:State.Name}' \
  --output table
-----------------------------------------------
|               DescribeInstances             |
+----------------------+----------+------------+
|          ID          |  State   |   Type     |
+----------------------+----------+------------+
|  i-0f7e10d91f7d745c8 |  running |  t3.micro  |
|  i-07a2652f332619565 |  running |  t3.small  |
|  i-0220ffef811ee3320 |  running |  t3.small  |
|  i-04af256bf16a376e6 |  running |  t3.small  |
|  i-012c9c1f00908aece |  running |  t3.small  |
|  i-07b13cdeee63debb4 |  running |  t3.small  |
+----------------------+----------+------------+
```

```
--region ap-southeast-2 \
--query 'AutoScalingGroups[0].{Desired:DesiredCapacity,Min:MinSize,Max:MaxSize,Instances:Instances[*].{ID:InstanceId,State:LifecycleState,Health:Hea
thStatus}}' \
--output json

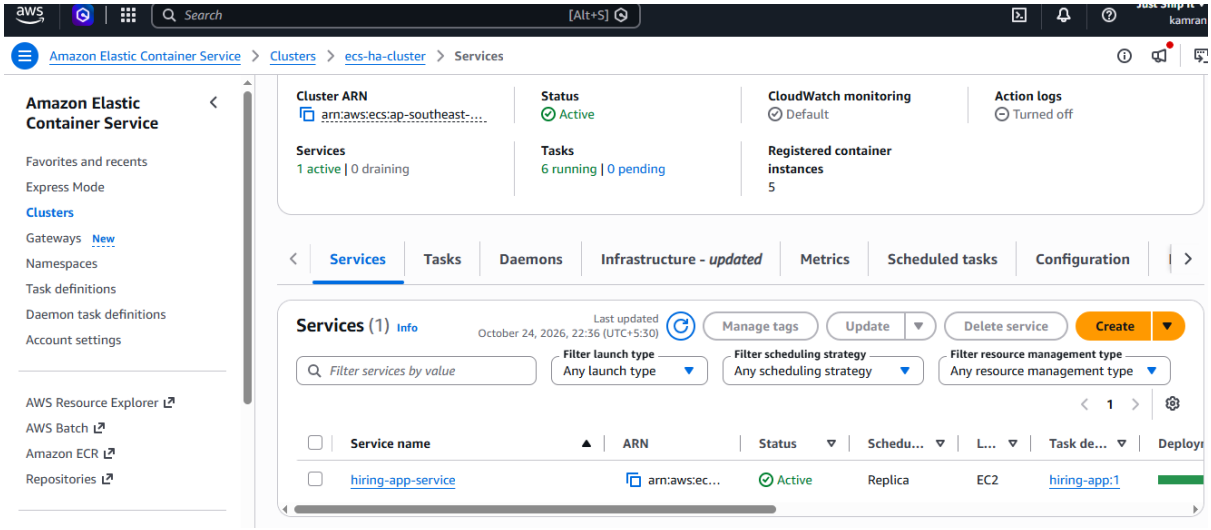
  "Desired": 6,
  "Min": 6,
  "Max": 9,
  "Instances": [
      {
          "ID": "i-012c9c1f00908aece",
          "State": "InService",
          "Health": "Healthy"
      },
      {
          "ID": "i-0220ffef811ee3320",
          "State": "InService",
          "Health": "Healthy"
      },
      {
          "ID": "i-04af256bf16a376e6",
          "State": "InService",
          "Health": "Healthy"
      },
      {
          "ID": "i-07a2652f332619565",
          "State": "InService",
          "Health": "Healthy"
```



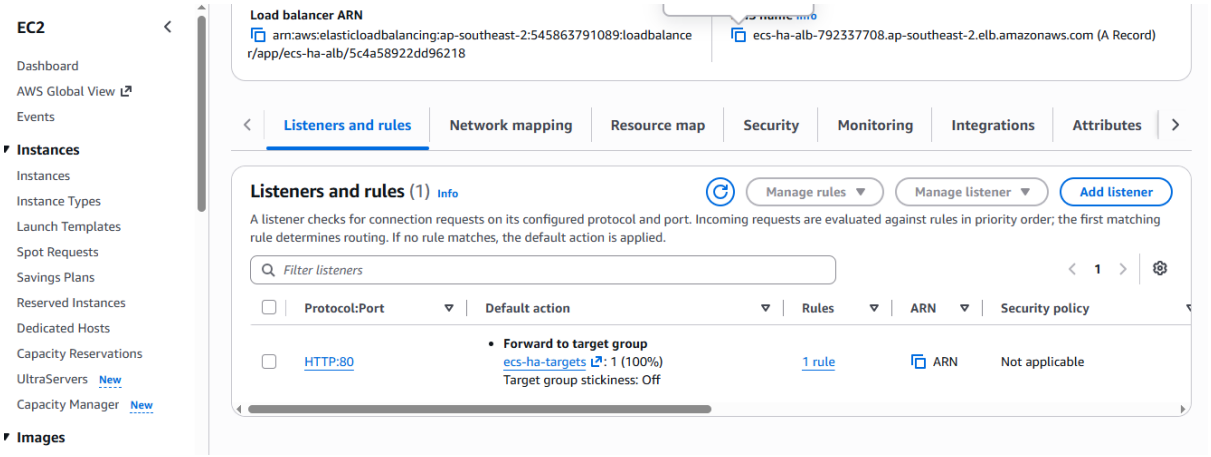
Target groups



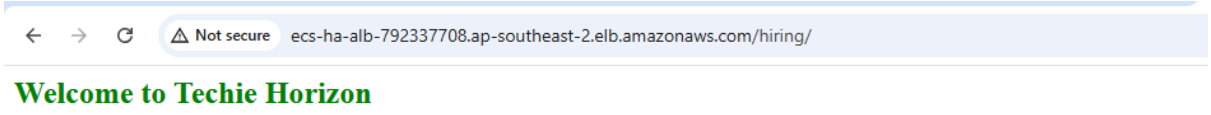
Esc cluster



Load balancer

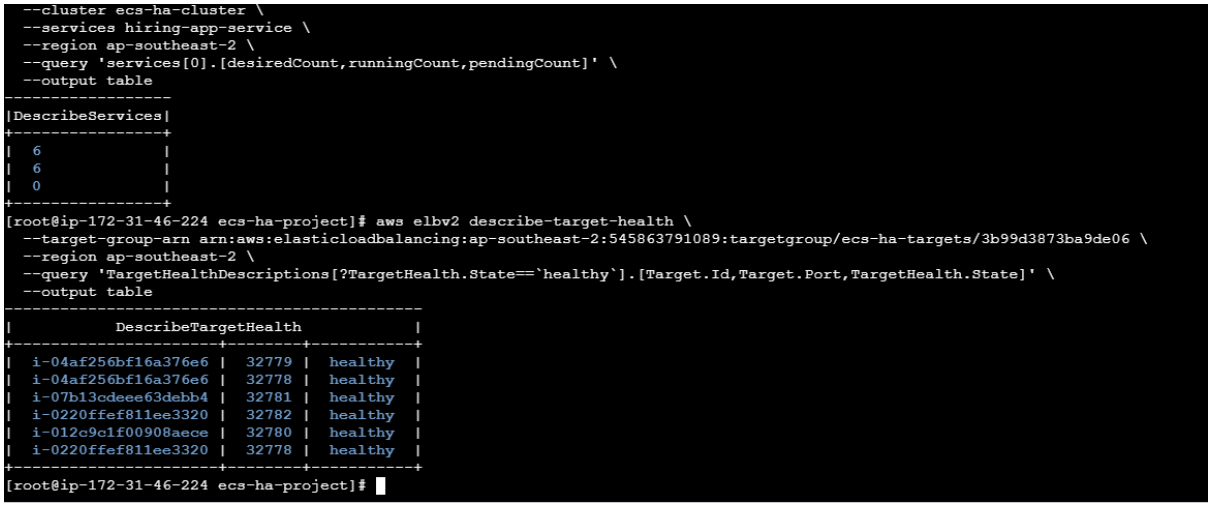


Application output



Welcome to Techie Horizon

Terminal output



AWS Permissions Configured – ECS High Availability

Objective: Provide the required AWS permissions and network access to deploy and manage the ECS cluster, load balancer, and application.

1. IAM Role and Permissions

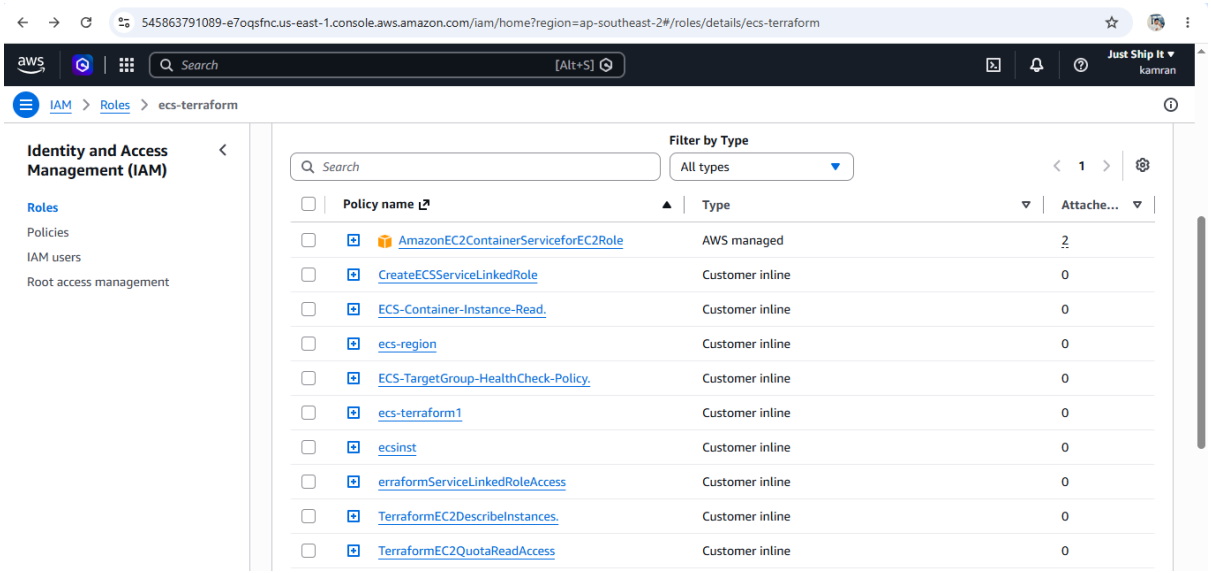
1. Used the IAM role `ecs-terraform` for AWS operations from the management EC2 instance.
2. Added permissions to describe ECS container instances.
3. Added permissions to modify the load balancer target group's health-check configuration.
4. Used the required ECS and Elastic Load Balancing permissions to configure and verify the deployment.

2. Security Group Rules

1. Allowed inbound traffic from the Application Load Balancer security group to the ECS instances.
2. Allowed TCP traffic on dynamic host ports `32768-65535`.
3. Restricted this dynamic-port access to the ALB security group rather than allowing traffic from all IP addresses.

3. Target Group Configuration

1. Configured the target group health-check path as `/hiring/`.
2. Verified that six targets were healthy.
3. Configured the application to use dynamic port mapping.



TerraformEC2QuotaReadAccess	Customer inline	0
TerraformECSAMIReadAccess.	Customer inline	0
TerraformECSAndSecurityGroupAc...	Customer inline	0
TerraformECSIAMAccess.	Customer inline	0
TerraformECSInstanceProfileAccess	Customer inline	0
TerraformECSLoadBalancerAndLa...	Customer inline	0
TerraformELBAndASGReadAccess	Customer inline	0
TerraformELBListenerReadAccess.	Customer inline	0